The Fly in the Ointment?

To me, this story of inflationary cosmology and time's arrow is lovely. From a wild and energetic realm of primordial chaos, there emerged an ultramicroscopic fluctuation of uniform inflaton field weighing far less than the limit for carry-on luggage. This initiated inflationary expansion, which set a direction to time's arrow, and the rest is history.

But in telling this story, we've made a pivotal assumption that's as yet unjustified. To assess the likelihood of inflation's being initiated, we've had to specify the characteristics of the preinflationary realm out of which inflationary expansion is supposed to have emerged. The particular realm we've envisioned—wild, chaotic, energetic—sounds reasonable, but delineating this intuitive description with mathematical precision proves challenging. Moreover, it is only a guess. The bottom line is that we don't know what conditions were like in the supposed preinflationary realm, in the fuzzy patch of Figure 10.3, and without that information we are unable to make a convincing assessment of the likelihood of inflation's initiating; any calculation of the likelihood depends sensitively on the assumptions we make.[6]

With this hole in our understanding, the most sensible summary is that inflation offers a powerful explanatory framework that bundles together seemingly disparate problems—the horizon problem, the flatness problem, the origin-of-structure problem, the low-entropy-of-the-early-universe problem—and offers a single solution that addresses them all. This feels right. But to go to the next step, we need a theory that can cope with the extreme conditions characteristic of the fuzzy patch— extremes of heat and colossal density—so that we will stand a chance of gaining sharp, unambiguous insight into the earliest moments of the cosmos.

As we will learn in the next chapter, this requires a theory that can overcome perhaps the greatest obstacle theoretical physics has faced during the last eighty years: a fundamental rift between general relativity and quantum mechanics. Many researchers believe that a relatively new approach called *superstring theory* may have accomplished this, but if superstring theory is right, the fabric of the cosmos is far stranger than almost anyone ever imagined.